

THE MOTIONS OF THE SKY

1. MOTIONS OF THE STARS

The Celestial Sphere

As seen from Earth, the sky can be thought of as a gigantic sphere that is centered on Earth to which all the individual objects seen in the sky are attached. This is called the *celestial sphere*. Points on the celestial sphere are directly above the points on Earth to which they correspond. The point on the celestial sphere directly above Earth's North Pole is called the *north celestial pole*. The point above Earth's South Pole is called the *south celestial pole*. The line directly above Earth's equator at all points is called the *celestial equator*.

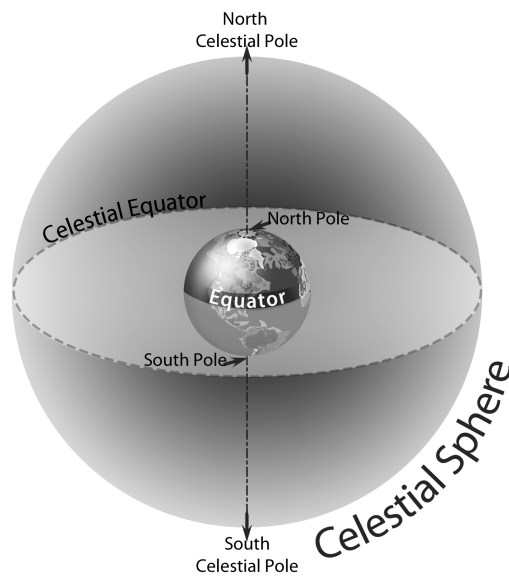


FIGURE 1 The Celestial Sphere.

Earth simply appears to be a circular disk centered on the observer that extends the same distance in all directions ending at a line called the *horizon*. The line where Earth ends and the sky begins. The sky is the shape of a hemisphere or dome that appears to center on the observer. The point in the sky directly above the observer is called the *zenith*. The compass points on the horizon: due north, south, east, and west, are collectively known as the *cardinal points*.

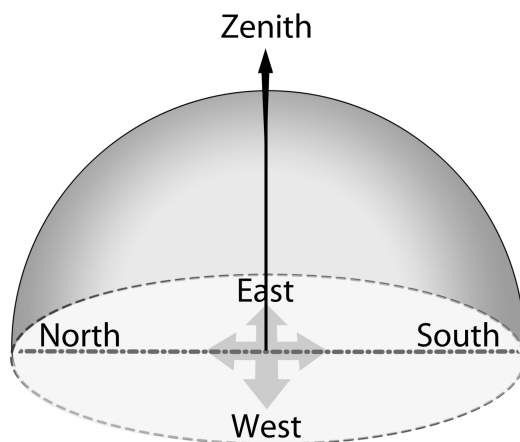


FIGURE 2 Earth and the sky's 'dome' from an observer's point of view.

What part of the celestial sphere an observer will see depends on where on Earth they are located. Note in Figure 2, that the angle from the zenith to the horizon is 90° , a right angle. An observer's zenith will always be directly above their location on Earth and they will be able to see any part of the celestial sphere above their horizon in their sky that is within 90° of their zenith. An observer *cannot* see anything that is below their horizon. For instance, referring back to Figure 1, an observer at Earth's North Pole will look straight up and see the north celestial pole *at* their local zenith. On their local horizon, 90° below their zenith, they will see the celestial equator. An observer at Earth's North Pole will see the entire northern half of the celestial sphere. All of the stars and other objects that are north of the celestial equator will be visible to them. However, the southern half of the sky is *not* visible from the North Pole. An observer there will *never* see objects south of the celestial equator. All the objects in the southern half of the sky are below their horizon. These are the objects seen by an observer at Earth's South Pole who will never see the objects in the northern half of the sky.

Observers at Earth's Equator will look straight up and see the celestial equator going through their zenith, making a line from due east to due west on the horizon. At their horizon they will see the celestial poles, the north celestial pole on their north horizon and the south celestial pole on their south horizon. Observers at Earth's equator see a "side" half of the celestial sphere and at one time or another are able to view *all* the objects on the celestial sphere.

The objects seen in the sky are mostly points of light that appear to be fixed to the celestial sphere that we call *stars*. Astronomers group the sky into eighty-eight sections they call *constellations*. Playing connect the dots with the stars of many constellations makes easily recognizable pictures in the sky. Figure 3 shows the region of the sky around the north celestial pole. The star that appears almost exactly at the north celestial pole, which would make it almost exactly above Earth's North Pole, is called *Polaris* or the *North Star*. There are common myths that Polaris is the brightest star in the sky or that it is the closest star to our Sun, but it is neither of these. What is special about Polaris is its location. Being above Earth's North Pole, Polaris will always indicate which way is north.



FIGURE 3 A common trick for finding the North Star, *Polaris*, is to use the two outer stars of the Big Dipper's bowl (bottom left) as *pointer* stars. An imaginary line drawn out of the Big Dipper's bowl along the line of the pointer stars will point to Polaris, the star at the end of the handle of the Little Dipper (upper right).

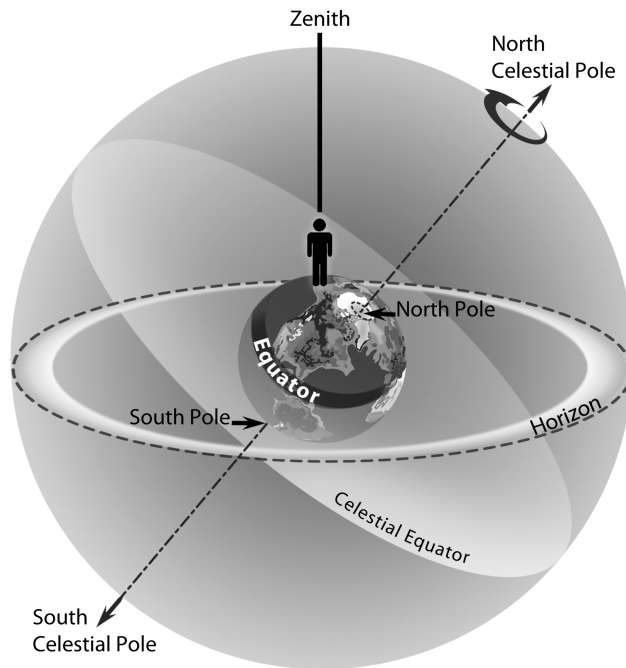


FIGURE 4 View of the celestial sphere to an observer located about halfway between the celestial equator and the north celestial pole.

An observer anywhere on Earth will always see half of the celestial sphere, but only those at the North and South Poles see exactly the northern and southern halves of the sky and only observers at Earth's equator see an exact "side" half. Much of the world's population lives about half way between Earth's equator and the North Pole. This region is called the middle-north latitudes. *Latitude* is the measurement of how far north or south a location is from the equator. As shown in Figure 4, an observer at a middle-north latitude location will look up and see a point in the sky that is about halfway between the celestial equator and the north celestial pole at their zenith. So, as seen in Figure 5, they will see the north celestial pole about halfway up from the horizon to the zenith in the northern part of the sky and the celestial equator about halfway up from the horizon to the zenith in the southern part of their sky.

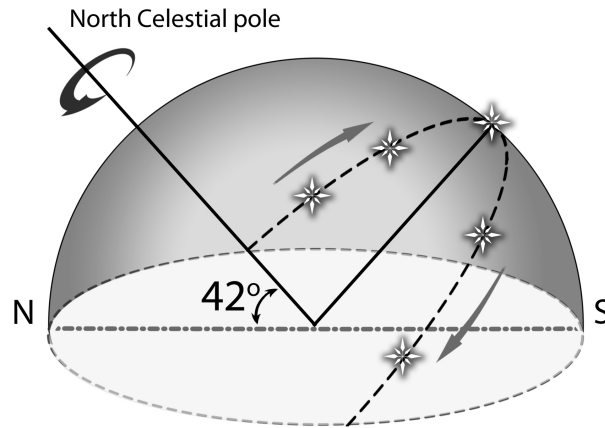


FIGURE 5 View of the sky for an observer at a middle-north latitude.

All the stars appear to move in circles around the north celestial pole from east to west as if they are attached to the celestial sphere. For stars near the north celestial pole, like those shown in Figure 3, their entire circle is above the horizon so they are stars that are visible every night. They are called *circumpolar* stars. Stars further from the north celestial pole will appear from below the horizon, or *rise* in the east; move across the southern part of the sky, reaching their highest angle above the horizon, or their highest *altitude*, when above the point due south on the horizon and then descend toward the west and disappear below the horizon, or *set* in the west. The arrows in Figure 5 point in the direction of the motion of the stars.

When a star crosses the meridian directly above due south, it is at its highest altitude of the day, this is called its *transit* altitude.

Celestial Coordinates

Just as latitude is a measurement of north and south on Earth, *longitude* is a measure of east and west on Earth. These are both shown in Figure 6. The equivalent measurements of latitude and longitude on the celestial sphere, also shown in Figure 6, are called declination and right ascension. *Declination* is the measurement of how far north or south an object in the sky is of the celestial equator while *right ascension*, like longitude on Earth, is a measurement east and west across the sky.

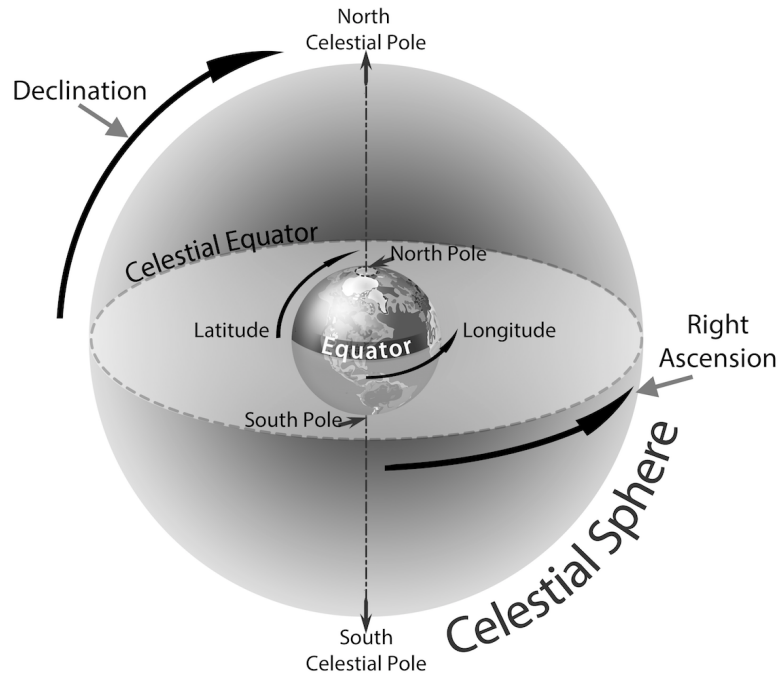


FIGURE 6 The Declination and Right Ascension of an object in the sky are equivalent to the Latitude and longitude of a location on Earth

2 MOTIONS OF THE SUN

Daily Motion

The daily motion of the Sun is similar to the motion of stars, see Figure 7. The Sun rises in the east and its altitude increases until it is directly above due south on the horizon and then its altitude decreases until it sets in the west. The line from due south to due north, passing through the zenith and dividing the sky into an eastern or rising half and a western or setting half is called the *meridian*. The time that the Sun crosses the meridian, when the Sun is in transit, is defined as local *noon*. The amount of time between two “noons” is our definition of the time measurement *day*. The hours before the Sun reaches the meridian are called *ante meridiem*, A.M. being the familiar abbreviation for the morning hours. The *post meridiem* hours, abbreviated as P.M., are the afternoon, the hours after the Sun has crossed the meridian.

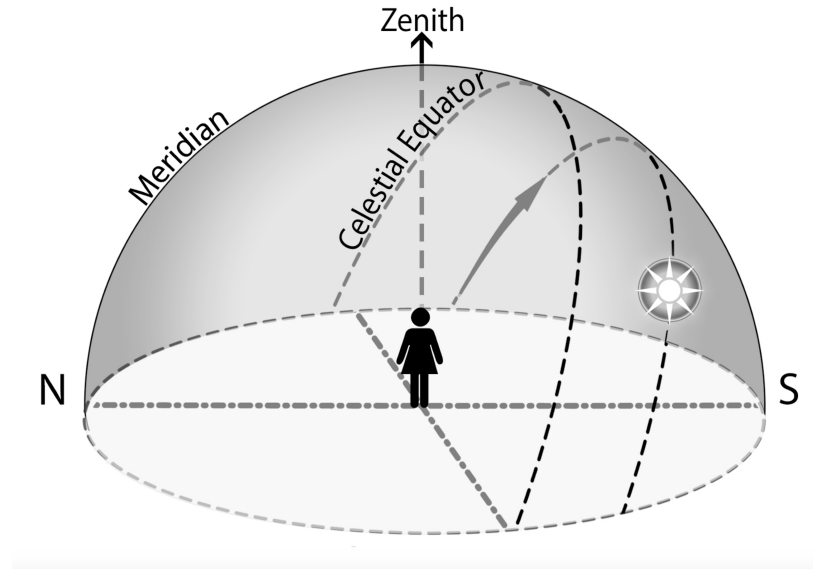


FIGURE 7 Daily motion of the Sun across the sky.

Annual Motion

The difference between the motion of the Sun and that of the stars is that a star's path through the sky will be the same every day, but the Sun's path is slightly different each day. If you observe the rising and setting positions and noon altitude of the Sun on any given day you will not notice much change in the next day or two. However, after about a week or more you *will* notice differences. The rising and setting positions along the horizon and noon altitude will change.

It is a common misconception that the Sun always rises exactly due east and always sets exactly due west. As can be seen Figure 9, when the Sun's rising and setting positions are at their furthest north of due east and west, the length of Sun's path through the sky is longest. This occurs about June 21, the *summer solstice*. When the Sun is high in the sky its rays strike the ground more directly and heat the ground more efficiently causing warmer weather. When the Sun is lower in the sky the rays are much less direct. The same amount of solar energy is spread out over a larger area so the ground is being heated less efficiently resulting in colder weather. See Figure 8.

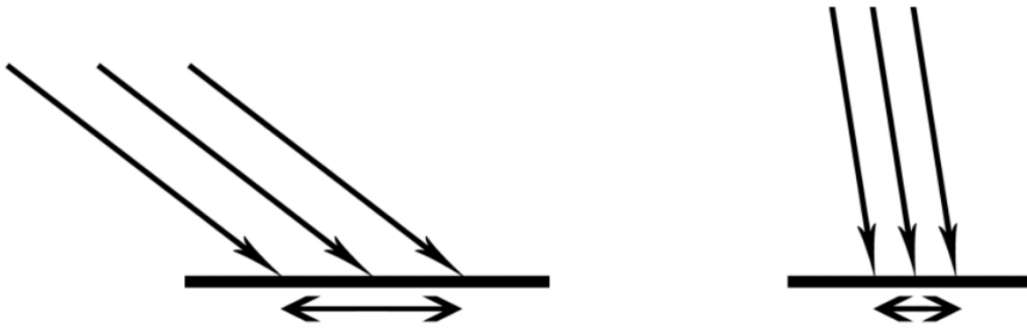


FIGURE 8 The Sun's rays striking the ground from a lower (left) and higher altitude.

Throughout the days of summer the Sun's path gradually moves southward and the noon altitude gets lower. Eventually, by about September 21, the rising and setting positions reach *exactly* due east and due west and the Sun is above the horizon and below the horizon (up and down) for twelve hours each. This is known as the *autumnal equinox*. Since the Sun's altitude is getting lower, the Sun is heating the ground less efficiently and weather in the fall becomes cooler.

The Sun's daily path will continue to move south, and the noon altitude will continue to decrease until about December 21. The day of the Sun's shortest path and its lowest noon altitude is called the *winter solstice*, the first day of winter. Winter is the time of year with the least daylight and coldest temperatures. After this, the Sun's rising and setting positions will start to move north again and noon altitudes will increase. There will be more daylight and the weather will get warmer. When the rising and setting positions return to due east and west the noon altitude will be the same as it was on the autumnal equinox and the Sun will again be up and down for twelve hours each. This is the *vernal equinox*, or first day of spring, that occurs about March 21.

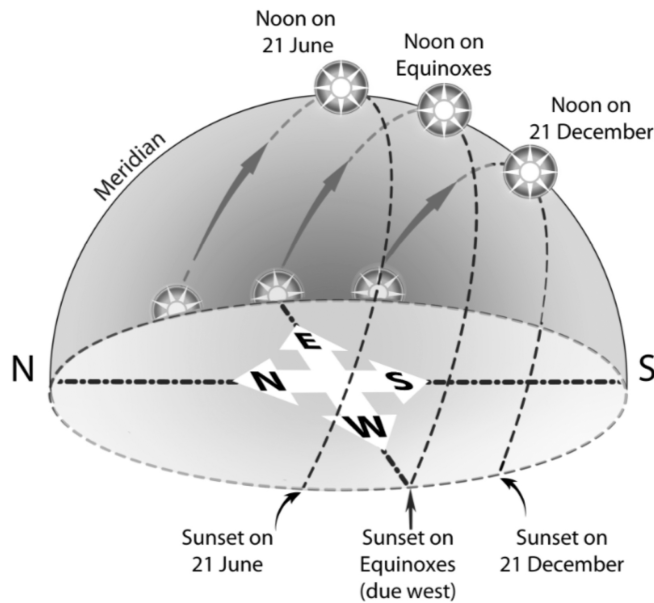


FIGURE 9 Differences in the Sun's daily path through the sky at different times of year.

The amount of time required for the Sun to completely cycle through its changing daily paths is our definition of the time measurement, *year*. So, like the day, the year is also a natural cycle of the Sun. A year is approximately $365\frac{1}{4}$ days. The extra $\frac{1}{4}$ day is the reason for leap years. Our calendar is 365 days, so every four years a day is added at the end of February to keep our calendar in sync with the Sun.

The Ecliptic

Each star has a fixed point on the celestial sphere like cities have fixed points on Earth. All stars all appear to move across the sky together once per day with the celestial sphere as if they are attached to their fixed points. Since the Sun's path is slightly different each day, the Sun has not just a point on the celestial sphere like each star, but its own path through the sky. A path extending all the way around the celestial sphere and tilted at an angle of $23\frac{1}{2}^\circ$ to the celestial equator, intersecting it at two points, is the Sun's path through the stars. This path is called the *ecliptic*.

On any given day as the celestial sphere appears to rotate the Sun moves with it. This results in the Sun's *daily* motion across the sky. By the next day, the Sun will have moved a small amount along the ecliptic and its observed daily path will be a little different. The Sun will appear to move completely around the ecliptic once in a year.

The points on the ecliptic are named for the northern hemisphere seasons. When the Sun is at its northernmost point it is high in the northern hemisphere and it is summer there. However, since the Sun appears high in the northern hemisphere sky, it will appear low in the southern hemisphere sky making it winter there. The reverse is true for the winter solstice. The southern hemisphere seasons are the opposite of those in the northern hemisphere.

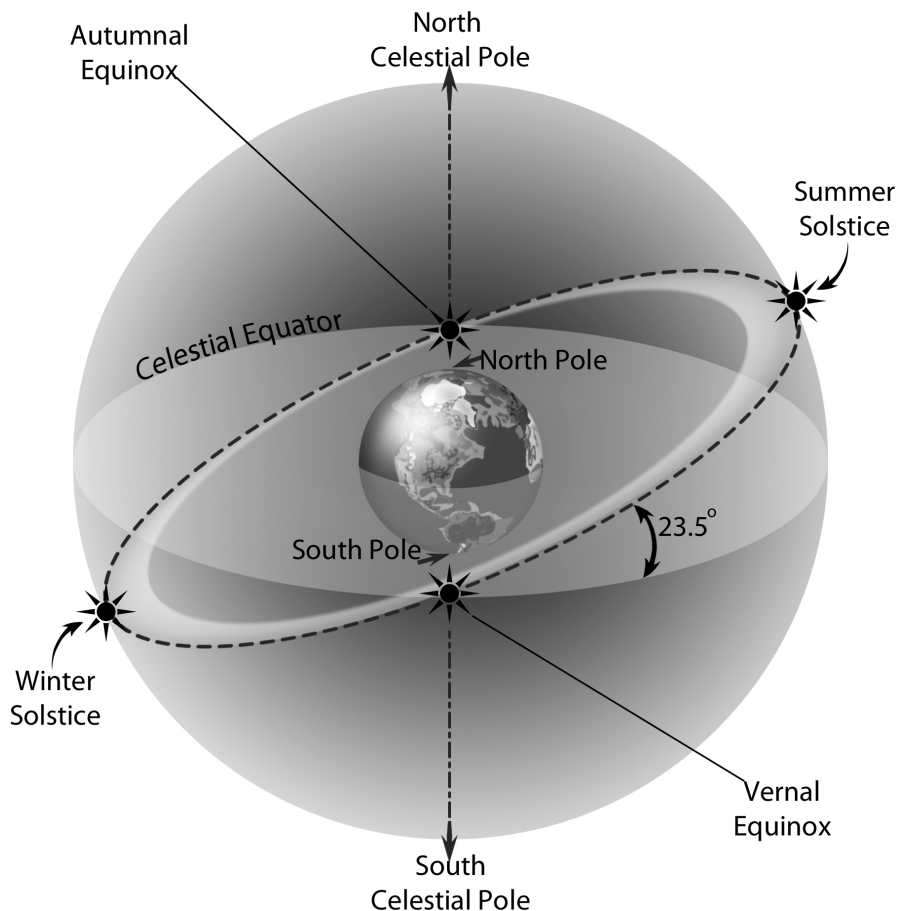


FIGURE 10 The ecliptic on the celestial sphere.

Because of the Sun's apparent motion on the ecliptic, its motion around the celestial sphere, the stars visible at different times of year change. When the Sun is at a given position on the ecliptic, stars and constellations near that location on the celestial sphere will *not* be seen because they are up *with* the Sun during the day. Those stars are all *still there* but we cannot see them because their light is washed out by the incredibly bright light of the Sun. The stars and constellations that *will* be seen are those on the opposite side of the celestial sphere from the Sun, the nighttime side. As the Sun moves around the ecliptic throughout the year different stars become visible at night. For this reason stars and constellations are *seasonal*. The night sky for a given season is that opposite the Sun's position on the celestial sphere at that time of year. Stars and constellations are visible at their highest altitudes of the year throughout the night during their season.

The stars that lie on and around the ecliptic are known as those of the *zodiac*. Historically, the zodiac had been divided into twelve constellations so the Sun spends *about* a month *in* each constellation. Among the eighty-eight constellations into which the sky has been divided in modern times there is actually a thirteenth constellation, called *Ophiuchus*, which also lies along the path of the ecliptic, so it should also be considered a member of the zodiac.

3 EXPLANATIONS FOR THE MOTIONS

Thus far, the observed motions of the stars and the Sun have been explained simply as they are seen from Earth, as if Earth is stationary and everything moves around it. The sky appears to spin or *rotate* around Earth once per day and the Sun appears to move *with* the sky each day. The Sun also appears to orbit or *revolve* around the Earth on the ecliptic once per year. This is an Earth-centered or *geocentric* perspective.

There is also a Sun-centered, or *heliocentric*, perspective. From this point of view daily motion of the sky is attributed to Earth rotating once per day from west to east *underneath* a unmoving or stationary sky. This causes the sky to appear to move in the opposite direction each day. This is just like riding a merry-go-round. Everything seems to be moving around you in the direction opposite your motion.

The Sun's annual motion can be attributed to Earth revolving in an orbit around the Sun once per year, making the Sun only *appear* to be moving along the ecliptic causing different stars to be visible at only certain times of the year.

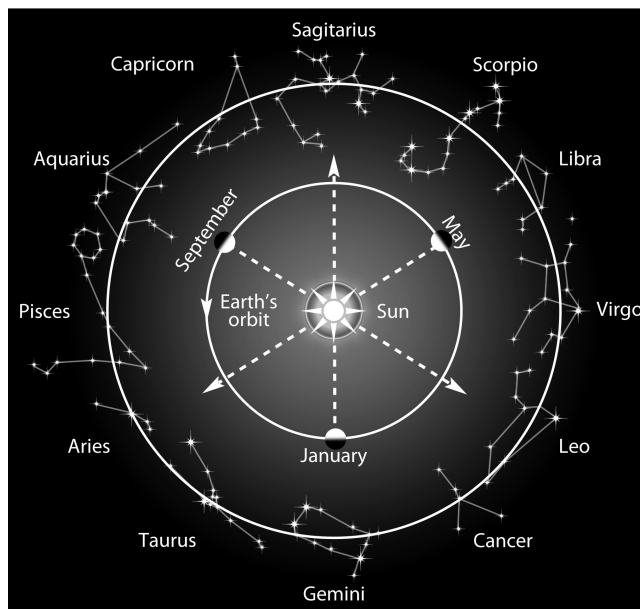


FIGURE 11 Earth's orbit around the Sun causes the Sun to appear to move through the stars and causes different stars to be visible at night at different times of the year.

From the heliocentric perspective, the seasonal changes in the Sun's daily path are explained by attributing the tilt not to the ecliptic but to Earth's rotational axis. As Earth revolves around the Sun once per year the Earth's axis always stays pointing in the same direction, toward the North Star, Polaris, causing different parts of Earth to receive different amounts of sunlight.

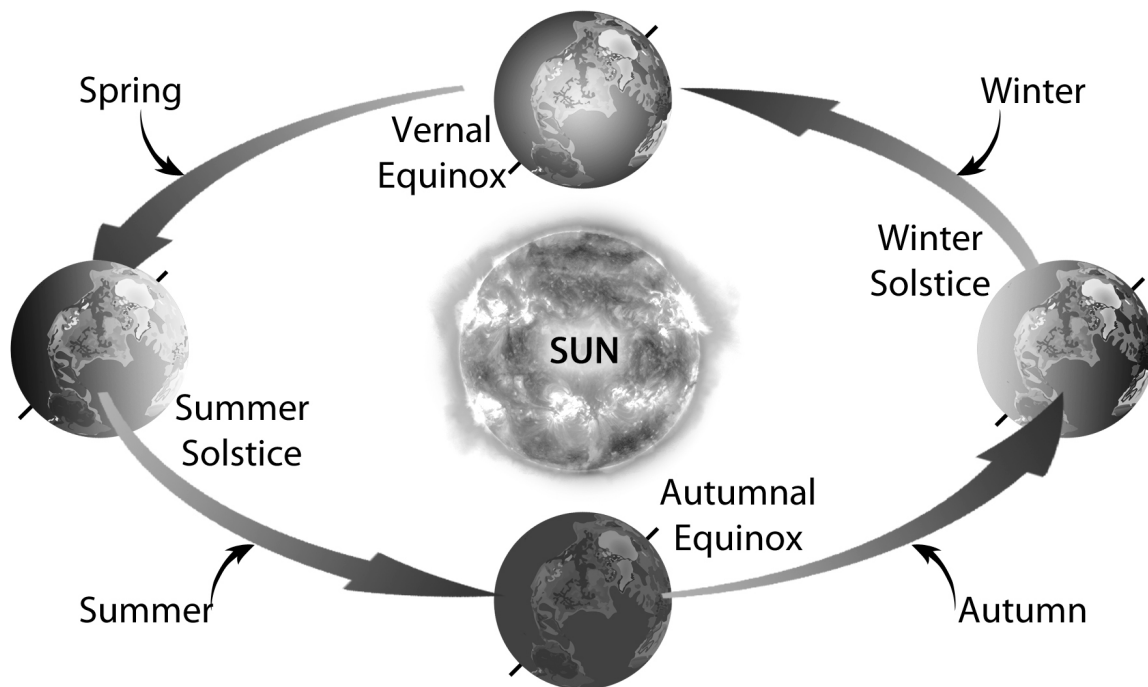


FIGURE 12 As Earth orbits the Sun its tilted rotational axis causes the seasons (sizes and distances in figure not to scale).

Note the angles of the Sun's rays, the arrows, in Figure 13. The North Pole is tilted toward them so the Sun's rays make larger angles with Earth's surface causing the Sun to appear higher in the sky so it is summer. Also note that as Earth rotates the northern hemisphere locations will spend more time in daylight and less in darkness. Note that the opposite is true of both the angle's of the Sun's rays and the time spent in daylight and darkness in the southern hemisphere where it is winter.

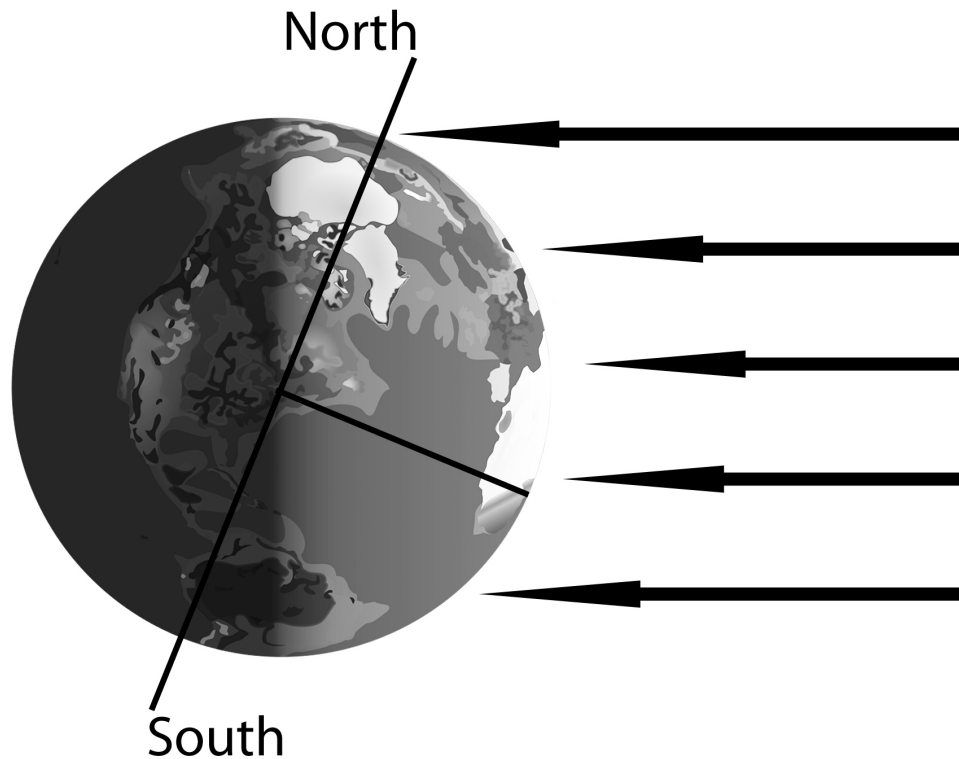


FIGURE 13 The Sun's rays during northern hemisphere summer.

4 MOTIONS OF THE MOON

The Moon goes through the most rapid changes in both appearance and position of any object in the sky. The variations in the amount of the Moon's surface that is visible from Earth are called *phases*.

Lunar Phases

The Moon cycles through its phases in a period of about twenty-nine and a half days, nearly thirty days, a *month*. The approximately thirty-day length of our calendar month is based on the Moon's cycle of phases.

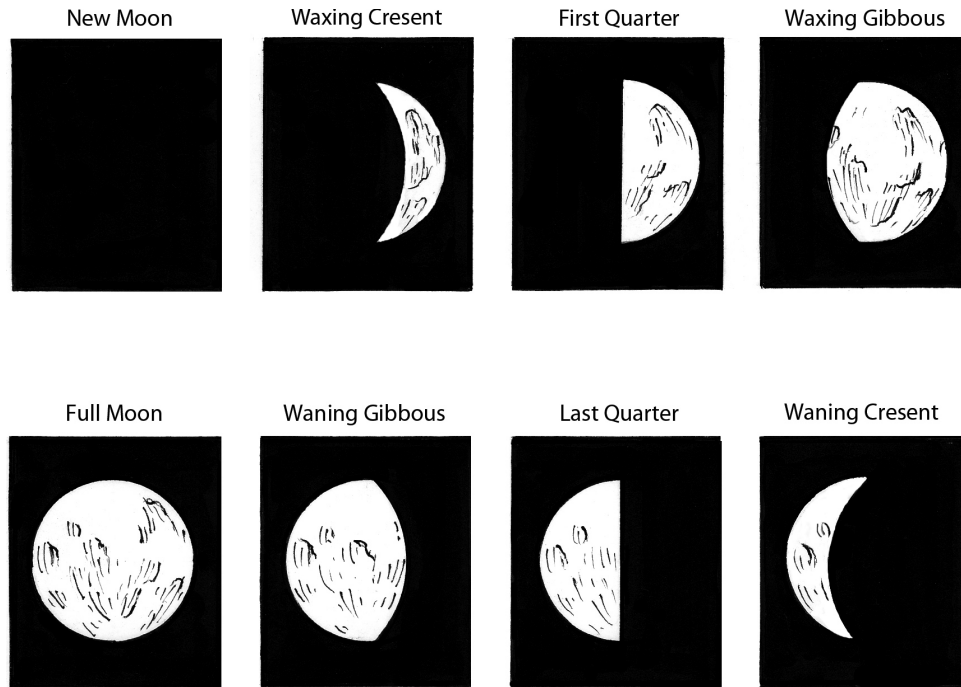


FIGURE 14 Lunar phases.

From the new to full phases the Moon is getting brighter so these are called *waxing* phases. The phases from full to new when the Moon is getting dimmer are called *waning* phases.

As the Moon orbits Earth the sides of both objects facing the Sun are illuminated while the sides opposite the Sun are dark. How the Moon will look from Earth in any given position in its orbit depends on how much of the side of the moon facing the Sun, the side receiving light, is also facing Earth.

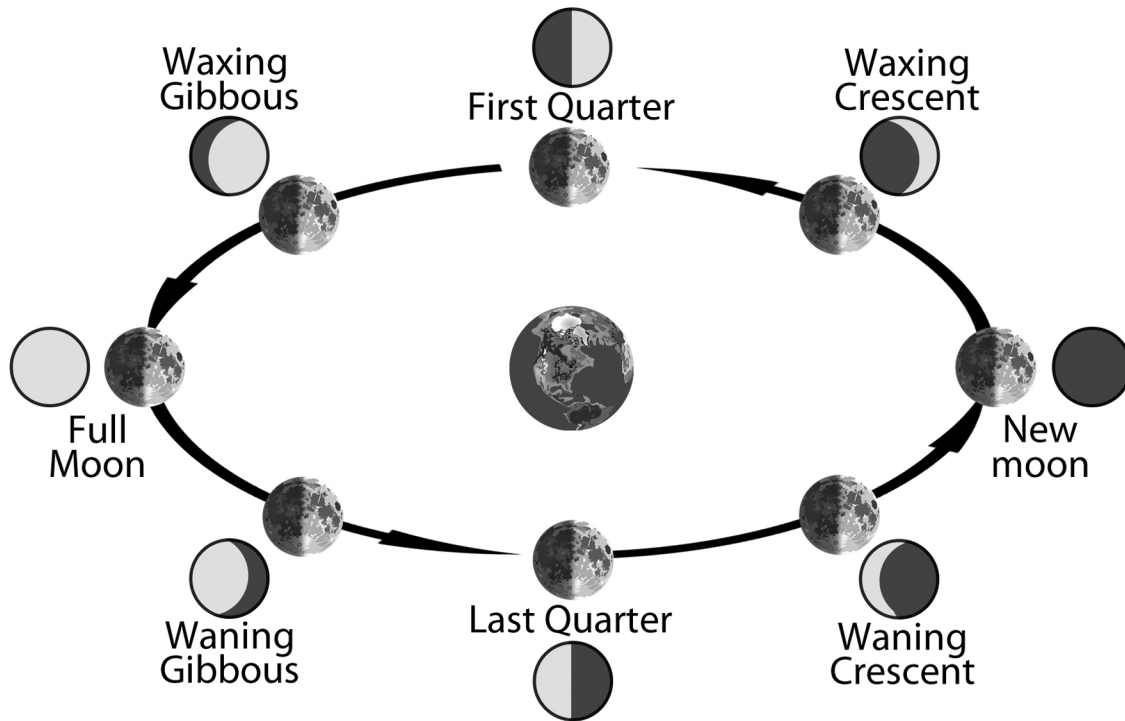


FIGURE 15 The phases of the Moon in the different positions in its orbit. The Sun's rays are coming from the right.

The best way to correctly visualize the phases of the moon is with an overhead view of the Moon in its orbit around Earth as shown in Figure 16. Consider yourself standing on the surface of Earth directly below the Moon's position in its orbit. Then, as shown in Figure 16, draw a line from your position on Earth to the Moon. Now draw another line, through the Moon separating the side of the moon that is facing you from the side that is not. This is also shown in Figure 16. *You can only see the side of the Moon that is facing you*, so look at the amounts of light and darkness that are facing you, then color in a circle with correct *amount* of darkness *on the side where you see it*. Remember that the picture you are looking at is a view from above, but you are using it to determine what you see from your position on Earth.

From the point on Earth directly below the Moon In Figure 16 you will see mostly darkness on the Moon's face and a little sliver of light on the left side. Coloring in a circle this way will give you a picture like the Crescent Moon shown in Figure 16.

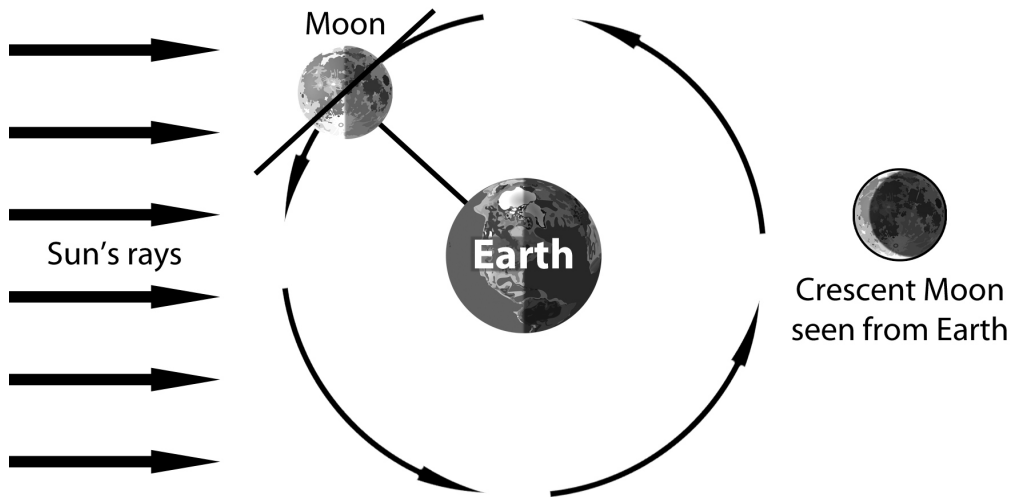


FIGURE 16 Using an overhead view of the Moon in its orbit around Earth to determine the phase of the Moon.

The times that the Moon will be visible are coordinated with its phases. It is noon at the point on the Earth directly below the Sun. In Figures 16 and 17 that is the point on the far left of the image of Earth. At the point on Earth opposite noon it is midnight. Viewed from above the North Pole, Earth rotates counter-clockwise. This defines the time of day at all other points on Earth. At the point halfway between noon and midnight, at the bottom of the image of Earth in the Figure 17 it is 6:00 P.M. . At the point halfway between midnight and noon, at the top of the image of Earth in the figures it is 6:00 A.M. These are also the times of sunset and sunrise, the positions where an observer would cross from being in daylight to darkness and vice-versa.

The time on the Earth-clock that the Moon is directly above the time that the Moon will be seen at its highest altitude for the day above due South on horizon. The moon will be visible from any point on Earth for the twelve hours that the point is facing the Moon, so the rising time of the Moon is *six* hours before its highest altitude for day and the setting time is *six* hours after that time or *twelve* hours after it rose. In Figure 17, the Moon is directly above 9:00 A.M. on the Earth-clock, so it rose at 3:00 A.M, will be at its highest altitude directly above due south at 9 AM and will set at 3:00 P.M.

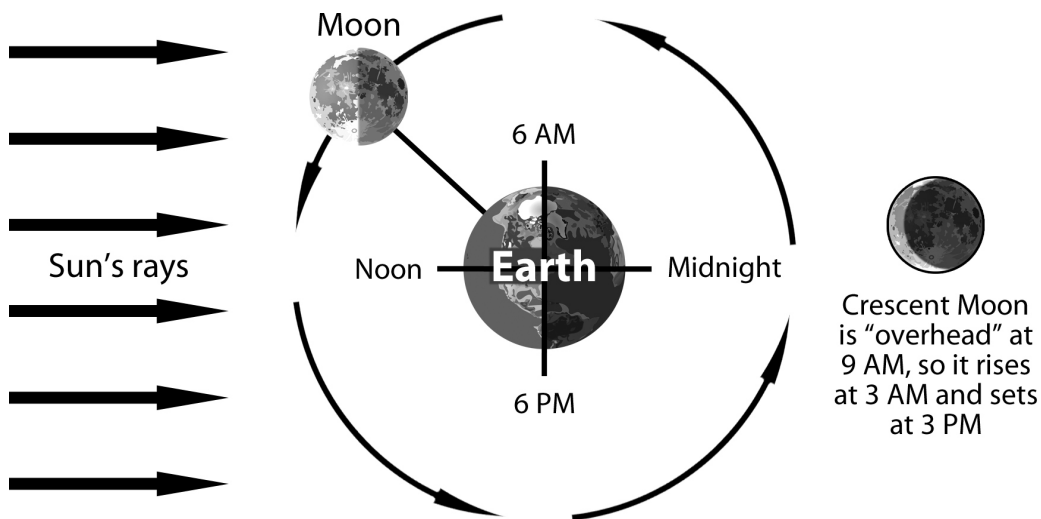


FIGURE 17 Using an overhead view of the Moon in its orbit around Earth to determine the rising and setting times for the Moon.



FIGURE 18 The near and far sides of the Moon.

The Far Side of the Moon

As the Moon orbits Earth the same side always faces towards Earth. There is a far side of the Moon that no observer on Earth ever sees. Astronauts orbiting the Moon in the Apollo 8 spacecraft first photographed the far side of the Moon in late 1968. The reason that the same side of the Moon always faces Earth is that the Moon rotates in exactly the same amount of time that it takes to revolve around Earth, a month. A common myth is that the Moon has a far side because it does not rotate.

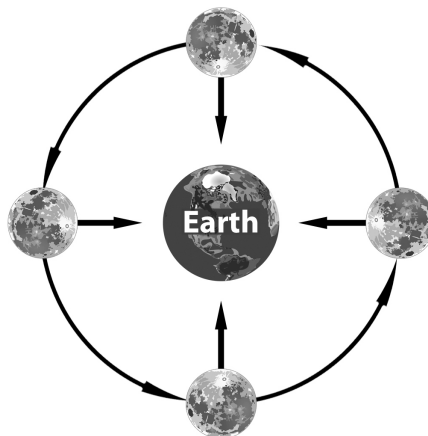


FIGURE 19 The same side of the Moon always faces Earth because the Moon rotates in the same amount of time that it takes to revolve around Earth. Note the changes in direction of the arrow coming from the moon in each position in its orbit.

5 ECLIPSES

From Earth, the Sun and the Moon both appear to be about the same angular size, about $\frac{1}{2}^\circ$. A fist held out at arm's length has an angular size of about 10° , a thumb about 2° , a finger 1° , and a pinky-finger about $\frac{1}{2}^\circ$. A pinky held out at arm's length can block out the Sun or the Moon. This is always true whether the Sun or the Moon are high overhead or close to the horizon. They both seem much larger when close to the horizon since there are objects near them to which they can be compared. This can be tested. No matter how large either object looks a pinky finger can still cover them both, try it!

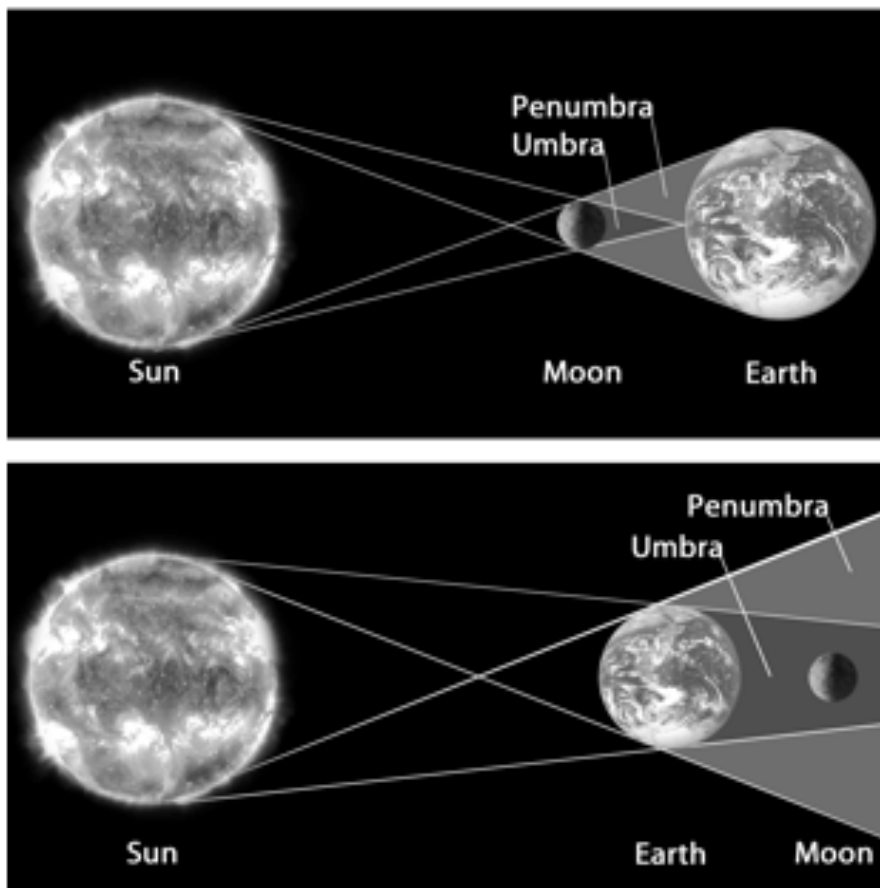


FIGURE 20 Solar and lunar eclipses.

The coincidence that the Sun and the Moon have the same angular size is the reason that events known as *eclipses* are so interesting. As indicated by the name, an *eclipse* can occur when the Moon is on or near the ecliptic. If the Moon is on the ecliptic when it is in between the Sun and Earth (in the new Moon phase) it will block out the view of the Sun from Earth. This is a *solar eclipse*. If Earth is between the Sun and the Moon (the full Moon phase) Earth's shadow will cover the Moon. This is a *lunar eclipse*.

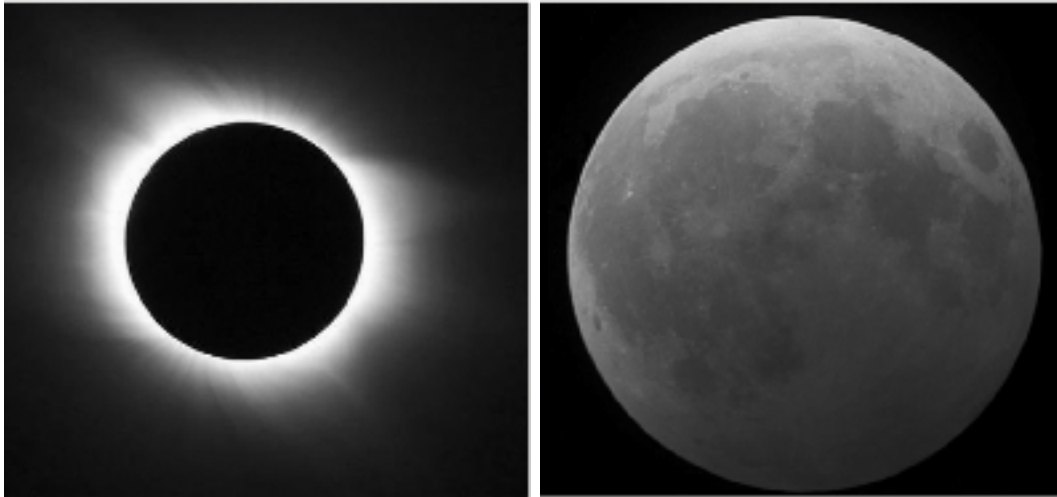


FIGURE 21 A total solar eclipse (left) and a total lunar eclipse.

Eclipses do not occur at every new and full Moon. This is because the Moon's orbit is tilted about 5° from the ecliptic. The ecliptic and the Moon's orbit are shown in geocentric perspective in Figure 22. The intersection points are called *nodes*. The line passing through Earth and connecting the nodes is known as the *line of nodes*. When the Sun and Moon are both on the line of nodes at the same time an eclipse will occur. The Sun traverses the ecliptic once per year so it will be on the line of nodes twice per year. If the Moon in its monthly orbit around Earth crosses a node *when the Sun is also at a node* there will be an eclipse. If the Sun is not at a node there will *not* be an eclipse. The times that the Sun is at or near a node are called *eclipse-seasons*. Since the Moon travels through the sky twelve times faster than the Sun it usually comes near enough to a node whenever the Sun is near one that eclipses occur about twice per year. If the Sun and Moon are at the same node a solar eclipse occurs. If they are at opposite nodes a lunar eclipse occurs. If the Sun and the Moon are not exactly lined up but are close a partial eclipse occurs.

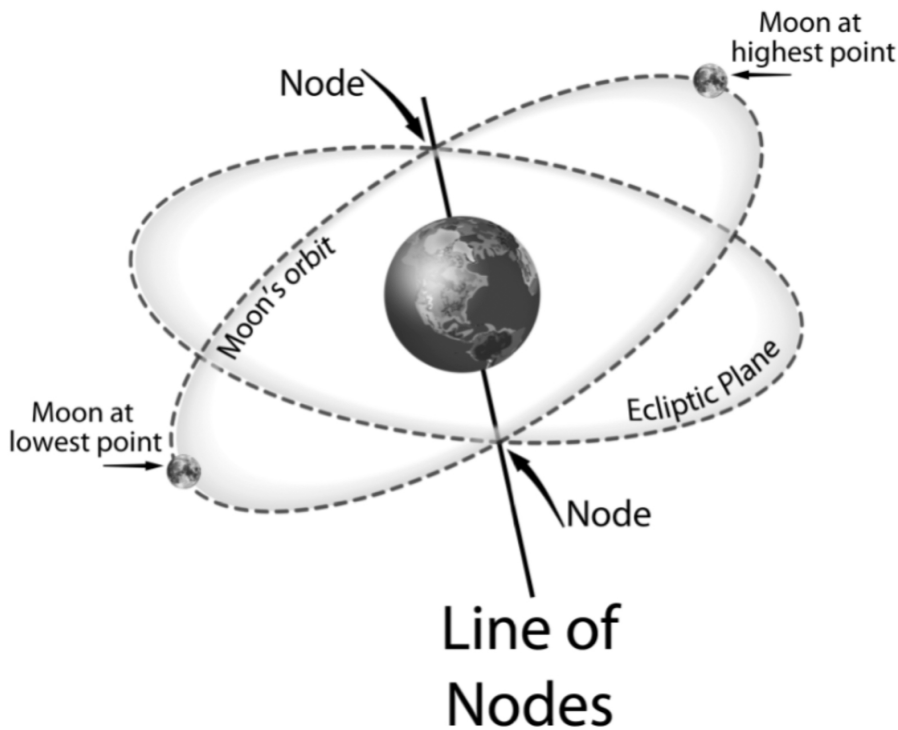


FIGURE 22 Eclipses occur when both the Moon and the Sun are on the line of nodes.

Referring back to Figure 20 shows that most people are likely to see more lunar eclipses in their lifetime than solar eclipses. During a lunar eclipse anyone on the entire nighttime side of Earth (the side facing away from the Sun) can see the full Moon eclipsed. During a solar eclipse only people in the specific position on Earth covered by the Moon's shadow will see the eclipse.

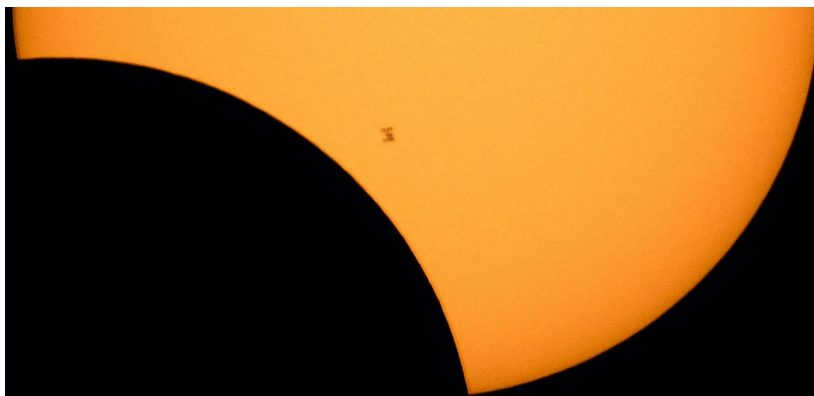


FIGURE 23 View of the International Space Station in front of the Sun during the 2017 eclipse.

Many people in the United States witnessed a solar eclipse that followed a path from the northwest to the southeastern parts of the country on August 21, 2017. Another that will be visible from the southwest to the northeast will occur on April 8, 2024.



FIGURE 24 Paths of the visibility of the 2017 and 2024 (not labeled) total solar eclipses across the United States

6. MOTIONS OF THE PLANETS

One year, the amount of time required for Earth to orbit the sun once, is Earth's sidereal period. The *sidereal period* for any planet is the time required for one orbit around the Sun.

A planet's *synodic period* is measured in reference to the Sun from Earth. It is the amount of time required, as viewed from Earth, for a planet to cycle through all possible configurations with the Sun.

The synodic period can be observed; the sidereal period, however, cannot be. It must be calculated from the Earth's sidereal period and the planet's observed synodic period. This is because a planet's orbit is centered on the Sun.

Figure 25 shows the synodic period of an *inferior planet*, a planet closer to the Sun than Earth (Mercury or Venus). When a planet lies along a line that passes through both the Earth and the Sun, it is said to be in *conjunction*. *Inferior conjunction* is when the planet is “below” the Sun or between the Sun and Earth. *Superior conjunction* is when the planet is “above” the Sun or the Sun is between the planet and Earth. A planet will be difficult to see when near either conjunction because of the brightness of the Sun.

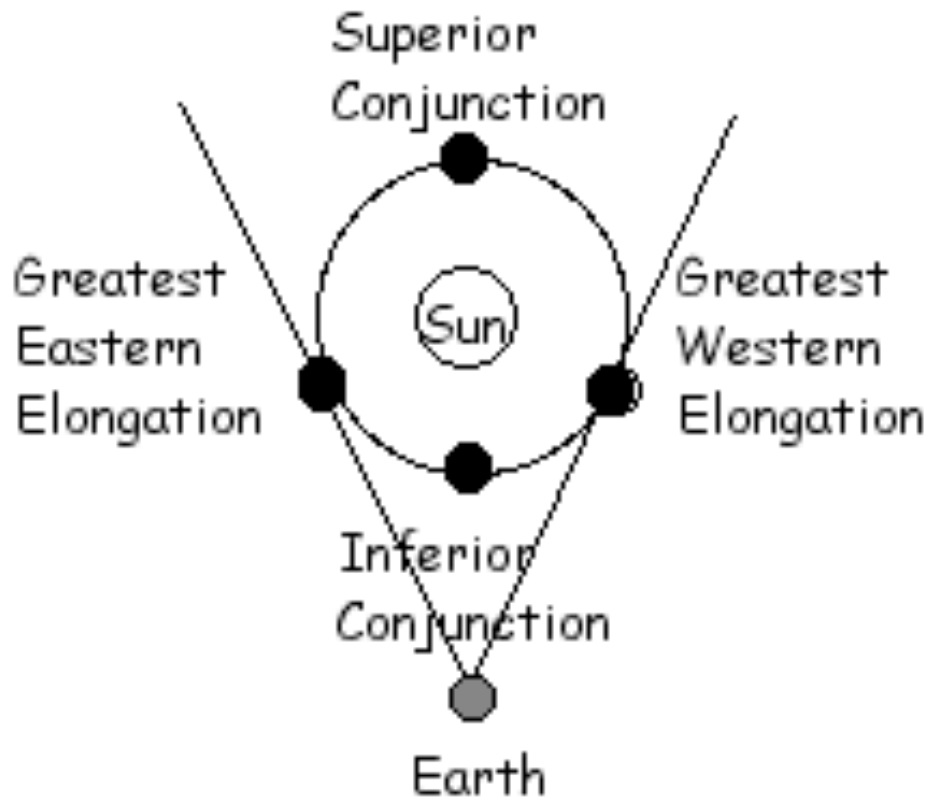


Figure 25 Configurations during the synodic period of an inferior planet

The apparent angle between an inferior planet and the Sun is called its *elongation*. Conjunctions are 0° elongation, and each planet has a greatest eastern and western elongation. Venus appears at most about 46° degrees from the Sun and Mercury at most about 28° . This is the position from which they can be best seen from Earth.

The configurations during the synodic period of a *superior planet*, a planet farther from the Sun than Earth are shown in Figure 26. *Superior Conjunction* occurs when the sun is in between the planet and the Earth. When the Earth is in between the planet and the Sun, it is said to be at *opposition*, because the planet is opposite the Sun in the sky. *Quadrature* is when the planet appears 90° from the Sun in the sky.

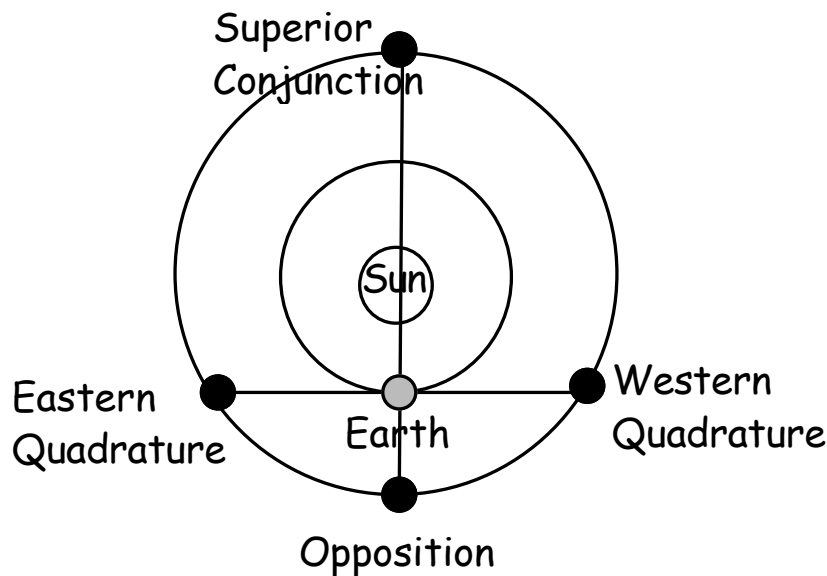


Figure 26 Configurations during the synodic period of a superior planet.

In order for a planet to be visible from Earth, two things must happen; the planet must be above the horizon and the Sun must be below the horizon. Determination of the times inferior and superior planets will be visible when in their different configuration will be left to activities and review questions.

REVIEW QUESTIONS

1. Describe the daily path of most stars. Where do they rise, reach their highest altitude, and then set?
2. Explain what is special about Polaris, the North Star.
3. About how long does the celestial sphere (the sky) appear to take to move around Earth once?
4. Describe the daily path of the Sun. Where does the Sun rise, reach its highest altitude, and then set?
5. At what time will the Sun reach its highest altitude for the day? How much time will have passed when the Sun reaches that altitude again?
6. Name three differences between the Sun's daily path in the summer and in the winter.

7. On what day(s) does the Sun rise exactly at due east and set exactly at due west?
8. How much time does the Sun take to move all the way around the ecliptic (completely through the zodiac) once?
9. During what month will Australia experience its shortest daylight period and lowest Sun altitudes of the year?
10. What are the two reasons that summer in the northern hemisphere is warmer than winter? What is the *cause* of these differences?
11. Explain how use of the term “Dark Side of the Moon” can cause confusion when referring to the Moon’s far side.
12. Refer to Figure 20. What type of eclipse would be observed from the Moon while the Earth is experiencing a lunar eclipse? What would be observed from the Moon while Earth is experiencing a solar eclipse?

13. Does a new Moon occur on the same day of every month on the calendar? Explain your answer.
14. Is the Moon ever visible during the day? Explain why or why not.
15. Name the motion responsible for the time measurements: day, week, month, and year.
16. Which configurations in Fig. 25 would be best for observing an inferior planet?
17. Which configurations in Fig. 25 would be the worst for observing an inferior planet?
18. Which configuration in Fig. 26 would be best for observing a superior planet ?
19. Which configuration in Fig. 26 would be the worst for observing a superior planet ?
20. What two conditions are necessary for a planet to be visible to observers on Earth?

